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YOLO-Based Character Detection and Sequence Recognition on Low-Contrast Faded Bangladeshi License Plates Using Synthetic Fading Augmentation

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Master's Thesis

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Abstract

Bangladeshi license plate recognition remains challenging under degraded visual conditions, particularly when fading and low contrast reduce character visibility. This study investigates synthetic fading augmentation to improve character detection and sequence recognition using lightweight YOLO-based models.

A dataset of 4,061 clean images, comprising 27 character and token classes, was divided into 3,480 training, 363 validation, and 218 test images. Two experimental families were conducted. The first compared a YOLOv8n baseline (Model A) with an augmented model (Model B) trained using the proposed BanglaPlateFade technique with an online augmentation probability of 0.60. The second compared YOLOv5su and YOLO11n models trained with and without synthetic fading. Performance was assessed using detection metrics, exact sequence accuracy, and character error rate (CER). Additionally, 67 real faded plate images were gathered from public sources, of which 64 had ground-truth labels for sequence-level evaluation.

On the clean test set, Model B achieved 95.70% mAP@0.50 and 71.73% mAP@0.50:0.95, compared with 94.13% and 70.49% for Model A. Recall improved from 89.18% to 92.49%, while exact sequence accuracy increased from 72.48% to 78.44% and CER decreased from 3.28% to 2.86%. On the real faded dataset, Model B achieved 67.19% exact sequence accuracy and 8.07% CER, outperforming Model A, which achieved 46.88% and 18.49%, respectively. In the additional experiments, synthetic fading improved YOLOv5s exact matches by 14.1 percentage points ($p = 0.0225$), although this result did not remain significant after Bonferroni correction. The Model A–Model B improvement was statistically significant ($p = 0.001$). These findings demonstrate the potential of synthetic fading augmentation for improving Bangladeshi license plate recognition under the evaluated conditions. However, the limited real faded dataset and lack of complete verified bounding-box annotations restrict conclusions about real-world detection robustness. Future work should evaluate larger annotated datasets and additional degradation conditions, including fog, rain, and nighttime environments.

Keywords: Bangladeshi license plate recognition, YOLO, synthetic fading augmentation, character detection, sequence recognition, character error rate.

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I . Introduction

This chapter introduces the background and motivation for robust Automatic License Plate Recognition (ALPR) on Bangladeshi plates, where physical fading and wear, combined with Bangla script and scarce annotated data, cause lightweight YOLO detectors to fail. It proposes synthetic fading augmentation as a low-cost remedy and asks whether it transfers to real faded plates. The chapter also states the problem, aim, objectives, research questions, contributions, and thesis structure.

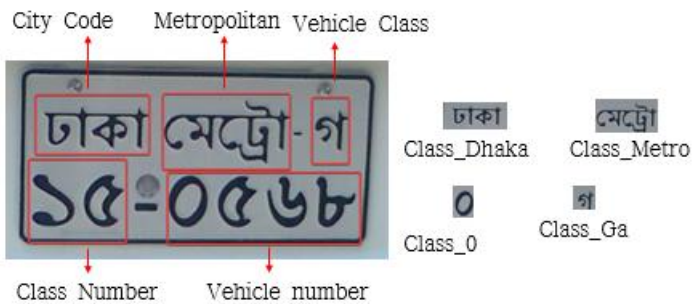


Figure 1: Structure and character classification of Bangladeshi License Plate

Automatic License Plate Recognition (ALPR) is the computer-vision task of locating a license plate within an image of a vehicle and transcribing its characters. Mature ALPR pipelines underpin intelligent transportation systems worldwide, including electronic toll collection, automated parking access, traffic-flow monitoring, stolen-vehicle detection, and electronic law enforcement. Because ALPR operates at the intersection of computer vision and public infrastructure, its reliability directly affects both operational efficiency and the fairness of automated enforcement.

Modern ALPR systems are overwhelmingly built on deep-learning object detectors. The YOLO (You Only Look Once) family of single-stage detectors is particularly popular for this task because it offers a favorable trade-off between inference speed and accuracy, enabling real-time operation on modest hardware. Recent lightweight variants — including YOLOv5s, YOLOv8s, and YOLO11n — reduce parameter counts and computational cost further, making deployment feasible at roadside units and embedded devices. A common ALPR design pattern, adopted in this thesis, treats character recognition as detection: each character class is detected individually, and the plate string is reconstructed by ordering the detected character boxes spatially. This unifies detection and recognition within a single model but introduces a sensitivity that is central to this study: if characters fade, the detector may miss them, and the reconstructed string then contains errors.

A persistent gap in the ALPR literature concerns robustness to physically degraded plates. License plates are painted metal exposed outdoors for years. Their characters progressively lose contrast as paint fades, oxidizes, and abrades — a degradation process distinct from, and in Bangladesh more prevalent than, transient weather effects such as fog or rain. Faded plates exhibit near-background character coloration, spatially non-uniform wear (some characters disappear while neighbors remain faintly visible), embossed shadows without ink, and rust or grime accumulation. These characteristics break the high-contrast dark-on-light assumption on which most detectors are trained. In this thesis, low-contrast fading denotes degradation in which the contrast between plate characters and the plate background is reduced, so that characters remain visible

mainly through embossed edges or shading and are hard to distinguish by color. It is distinct from blur, occlusion, and weather effects.

Bangladesh presents a compounded challenge. Bangladeshi vehicle plates pair Bangla script (with conjunct characters and a headline stroke) with Bengali numerals and Latin letters, following the format Division [Metro] Letter – Number (e.g., ঢাকা মেট্রো-গ ৩১-৯৯৫৭). Models pretrained on Latin-script plate datasets may therefore not transfer directly. At the same time, publicly available annotated Bangladeshi plate data are scarce, and data depicting degraded plates are scarcer still: collecting and annotating large numbers of real faded plates is slow, and the resulting collections are small — in this study, 67 images, of which 64 carry verified ground-truth transcriptions.

Synthetic image augmentation offers a way out of this data scarcity. If the visual process of fading can be approximated computationally — blending character ink toward the plate background color, randomizing wear per character, and superimposing rust and grime textures — then large quantities of faded training data can be generated from an existing clean corpus at no annotation cost, since annotations are preserved by construction. Whether such synthetic fading transfers to real faded plates, however, is an empirical question and the motivating question of this thesis.

1. Problem Statement

Object detectors that perform well on clean images can lose accuracy on degraded ones: a benchmark study found that standard detectors fell to as little

as 30–60% of their original performance on corrupted images [1]. Photographs of real vehicles can contain faded, faint, or partially obscured plate characters, so license plate recognition systems must handle such images.

For Bangladeshi plates, a character-level detector must both detect each character and reconstruct the complete plate number in the correct order. A recent Bangladeshi system that recognizes plate characters with YOLO reports full-number recognition accuracy alongside detection metrics [2]. Missing detections, duplicate bounding boxes, false positives, and incorrect classifications can each corrupt the sequence, and a single error makes the whole number wrong. In the present results, duplicate boxes affected several plates (Section 4.2).

Training with synthetically degraded images may help a detector keep working under reduced image quality. Synthetic images have been used to train license plate recognition networks successfully, provided the training images vary enough in the factors that control plate appearance [3] and a Bangla plate system used synthetic images to recognize plate classes under unfavorable and complicated conditions [4]. Robustness is commonly studied with benchmarks built from corrupted versions of standard datasets [5], [1]. However, improvement on synthetic images does not by itself show improvement on real faded plates. The mismatch between simulated and real data is known as the reality gap, and domain randomization was proposed to bridge it [6]. Separate evaluation on real faded images is therefore needed.

Detection metrics such as precision, recall, and mAP are computed against

annotated ground-truth boxes, as in the standard PASCAL VOC evaluation procedure, which supplies annotated images and fixed evaluation rules [7]. The real faded dataset contains 67 cropped plate images, of which 64 have labeled plate strings. It has no verified character bounding boxes, so detection mAP cannot be computed for it. The 64 labeled images can still support a recognition evaluation, provided predictions and ground-truth sequences are processed in the same way.

The central problem addressed in this research is therefore whether training lightweight Bangladeshi license plate character detectors with synthetic fading improves recognition on real faded plates, evaluated consistently across clean, synthetically faded, and real faded conditions.

2. Research Objectives

The main aim of this research is to evaluate the robustness of lightweight deep-learning-based detectors for Bangladeshi license plate character detection under clean and degraded image conditions and to investigate whether training with synthetic fading improves performance on real faded license plates. The following specific objectives are defined:

- To establish a baseline performance measurement for lightweight object detectors on clean Bangladeshi license plate images.
- To generate a synthetically faded dataset from available clean license plate images and evaluate detector performance under this degradation condition.

- To compare models trained on clean data with models trained using clean and synthetically faded data.
- To evaluate the generalization of the trained models on a separate collection of real faded license plate images.
- To measure complete-sequence recognition accuracy and character-level error rates on real faded plates with available ground-truth text.
- To conduct consistent model comparisons using appropriate evaluation metrics and, where the data and experimental design permit, paired statistical tests.
- To identify the limitations of synthetic degradation, annotation availability, and post-processing in the evaluation of real-world license plate recognition.

3. Research Questions

RQ1. How does the detection performance of lightweight YOLO detectors change between clean and synthetically faded Bangladeshi license plate images?

RQ2. Does training with synthetic fading improve detection on synthetically faded images compared with training on clean images alone, without reducing clean-image performance?

RQ3. To what extent does the benefit transfer to a separate collection of real faded plates, measured by exact registration-number accuracy and character error rate?

RQ4. Are the observed differences between models statistically supported by the available paired test data?

RQ5. How do post-processing choices (duplicate suppression and confidence threshold) affect recognition on real faded plates?

4. Contributions of This Thesis

The major contributions of this thesis are as follows:

I. A comparative robustness evaluation. The study compares lightweight detectors across clean and synthetically faded Bangladeshi license plate images rather than relying exclusively on clean-image performance.

II. An evaluation of synthetic-fading training. The research examines whether including synthetic fading in the training process changes detection performance under degraded conditions.

III. A separate real-faded recognition assessment. The study evaluates complete plate-sequence recognition on a real faded dataset with available ground-truth text, providing a separate test of generalization beyond the synthetic evaluation set.

IV. Multiple complementary evaluation metrics. The study considers object detection metrics alongside sequence-level and character-level measures to show how detection errors affect the final plate number.

V. An emphasis on reproducible evaluation. The research identifies the importance of checkpoint provenance, consistent post-processing, reliable annotations, and per-image paired comparisons when drawing conclusions from model experiments.

5. Thesis Organization

This thesis is organized into five sections as follows.

Section Ⅰ introduces the research problem, motivation, and contributions. It discusses the challenges of physical degradation model underlying the proposed method, and states the specific research objectives.

Section Ⅱ reviews related work of automatic license plate recognition, deep-learning-based object detection, lightweight YOLO models, image degradation, synthetic data augmentation, and the evaluation of recognition robustness. It identifies the research gap addressed by this study.

Section Ⅲ research methodology describes the datasets, preprocessing, synthetic-fading procedure, model configurations, training strategy, evaluation metrics, statistical methods, and reproducibility procedures.

Section Ⅳ experimental results and discussion presents and interprets all testing data, model comparisons, and statistical analyses. It evaluates these findings against research questions and details limitations like dataset size and synthetic degradation.

Section Ⅴ conclusion and future work summarizes the main findings, addresses the research objectives, and proposes directions for improving robustness and evaluating license plate recognition under more realistic adverse-weather conditions.

II. Literature Review

1. Automatic License Plate Recognition: Overview

Automatic License Plate Recognition (ALPR) is the task of locating a license plate in an image of a vehicle and transcribing its character sequence. Because license plates are rigid, standardized, and legally required to be visually distinctive, they function as a natural "visual license" for vehicles — a property that ALPR systems exploit across intelligent transportation applications, including electronic toll collection, parking access control, traffic monitoring, and automated enforcement. A mature ALPR pipeline typically decomposes into three stages: plate localization (finding the plate region), character segmentation or detection (isolating individual characters), and character recognition (classifying each character and ordering it into a string). Early systems performed these stages with hand-engineered features — edge detectors, color models, and projection profiles — and classical classifiers such as template matching and support vector machines. While such systems could reach high reported accuracy under controlled conditions, their reliance on high contrast, fixed viewpoints, and clean plates made them fragile in unconstrained deployment.

The deep-learning era fundamentally changed this. Convolutional neural networks (CNNs) replaced hand-crafted features, and the introduction of general-purpose object detectors — R-CNN and its descendants [8], [9] and the YOLO family of single-stage detectors [10]— enabled researchers to treat both plate localization and character recognition as unified detection problems. Two influential works shaped the modern detection-based paradigm that this thesis

adopts. [11] demonstrated segmentation- and annotation-free plate recognition in which characters are localized and classified directly by a deep detector, with a failure-identification mechanism rejecting low-confidence reads. [12] Built a complete layout-independent ALPR system on the YOLO detector, combining plate detection, layout classification, and character recognition with post-processing rules, achieving an average end-to-end recognition rate of 96.9% across eight public datasets — outperforming both prior academic systems and commercial engines on several of them.

2. Plate Detection Versus Character Detection

A useful distinction in the ALPR literature separates plate detection (one bounding box per plate) from character detection (one bounding box per character). Plate detection is the more widely studied problem and is now considered nearly solved for high-quality imagery: modern detectors routinely exceed 95% mean average precision on standard benchmarks. Character detection is harder and less standardized. Characters are small objects occupying few pixels; they repeat identical shapes across plates (e.g., the digit ᄂᆞ appears in most plates), creating systematic class imbalance; and their spatial arrangement carries the sequence semantics, so a detector that misses, duplicates, or misorders characters corrupts the final string even when per-character accuracy is high.



Figure 2: License plate Detection

The detection-based character approach has two practical attractions. First, it requires only bounding-box annotations of the kind produced by standard tools, avoiding the segmentation masks needed by pixel-level methods.



Figure 3: Character bounding boxes

Second, recognition becomes layout-independent: the same detector trained on character boxes can read plates of different formats by ordering boxes spatially,

without a format-specific segmentation module [12]. The cost, as [11] observed, is sensitivity to image quality: when characters fade toward the background, character-level recall degrades discontinuously, and sequence accuracy collapses with it. This trade-off annotation simplicity and layout independence on one side, degradation sensitivity on the other — motivates the robustness focus of this thesis.

3. Character Recognition and Sequence Reconstruction

Sequence reconstruction from detected characters is a distinct sub-problem with three practical components. First, detected character boxes must be ordered (typically left-to-right by center-x coordinate, with line grouping for two-line plates). Second, duplicate suppression is required because detectors frequently emit multiple overlapping boxes for the same character; naive argmax-per-position decoding then produces spurious repeated characters in the output string. Third, normalization must reconcile formatting differences between prediction and ground truth — spacing, dash placement, and the choice of Bengali versus Arabic-Indic digit glyphs — without erasing semantically meaningful characters such as leading zeros.

An alternative lineage avoids detection-based reconstruction entirely by treating OCR as sequence generation. Recurrent architectures with connectionist temporal classification [13] and, more recently, vision-encoder–decoder transformers have been applied to cropped plate images. In the Bangla context, a recent framework paired YOLOv8-based localization with a ViT–BanglaBERT vision–language OCR

stage, reporting a character error rate of 0.1323 on Bangla plate text reporting a character error rate of 0.1323 on Bangla plate text while noting that the underlying localization training data were predominantly daytime imagery [14]. Sequence-generation recognition is a genuine competitor to detection-based reconstruction, and this thesis treats it as future work (see Section 6); the present study deliberately retains the detection-based design because robustness of the detector itself, under fading, is the object of study.

4. Deep Learning Object Detection

Two-stage detectors such as Faster R-CNN [9] generate region proposals before classification and remain strong accuracy references, but their two-pass architecture is expensive for real-time use. Single-stage detectors — YOLO [10], SSD, and RetinaNet [15], via focal loss — predict boxes and classes in a single forward pass, trading a modest accuracy penalty for an order-of-magnitude speed advantage. Feature pyramid networks [15] addressed the small-object weakness of early single-stage detectors by building multi-scale feature hierarchies — a capability of direct relevance to character detection, where characters occupy tens of pixels. Transformer-based detectors such as DETR [16], ECCV removed hand-designed anchors and NMS at further computational cost; for lightweight deployment scenarios, however, convolutional single-stage detectors remain the dominant engineering choice.

5. The YOLO Family and Lightweight Detectors

The YOLO family evolved rapidly. YOLOv1 [10] introduced unified detection; YOLOv2/YOLO9000 [17] added anchor boxes and multi-scale training; YOLOv3 [18] introduced multi-scale prediction heads at three resolutions; YOLOv4 [19] systematized the "bag of freebies" training tricks; YOLOv5 [20] (Ultralytics, software release without a formal paper) repackaged the recipe in PyTorch and became the de facto community standard; YOLOv7 [21] and the anchor-free YOLOv8 [22] refined the architecture with decoupled heads and distribution-focal-loss regression; and YOLO11 targets improved accuracy–parameter efficiency for edge deployment.

Two properties of this lineage matter for the present study. First, the small-model variants used here (YOLO11n, YOLOv8n, and YOLOv5s) have checkpoints of roughly 5 to 18 MB, which makes them candidates for roadside deployment in resource-constrained settings, although inference speed was not measured in this study. Second, successive YOLO releases differ less in fundamental architecture than in training recipes and head design; a robustness benchmark across three family members (YOLOv5s, YOLOv8n, and YOLO11n) is therefore more informative than a single-model study, because conclusions about degradation and augmentation can be checked for family-level generality rather than single-checkpoint idiosyncrasy. This cross-family design follows comparative ALPR studies that evaluate several YOLO generations on the same data [23].

6. Robustness Under Visual Degradation

A substantial body of work establishes that deep detectors are brittle under

distribution shift. Michaelis [1] systematically benchmarked object detection under corruptions — noise, blur, weather, digital artifacts — at multiple severities and showed consistent, sometimes catastrophic, mAP degradation that generic ImageNet training does not prevent. For ALPR specifically, the most relevant degradations are contrast loss, blur, and illumination change, and the most studied real-world correlate is adverse weather. The canonical restoration line of work — dark-channel-prior dehazing [1] and its learned descendants — treats weather degradation as an image-formation problem to be inverted before recognition.

The degradation addressed in this thesis is fundamentally different in kind. Fading of painted plates is material and irreversible: ink chemically or mechanically lost, per-character and non-uniform, often accompanied by rust and grime. No restoration algorithm can recover information that is physically absent from the plate surface — a faded plate photographed perfectly is still a faded plate. This distinguishes the present problem from fog (a transient atmospheric veil that restoration can largely remove) and aligns it more closely with the low-visibility character-degradation problem. Notably, prior Bangla work targeting weather robustness has simulated fog by augmentation or dehazing pre-processing [24] but among the studies reviewed here, none evaluated whether fading-trained models transfer to physically faded plates.

7. Synthetic Data Augmentation and Domain Generalization

Synthetic augmentation as a remedy for data scarcity has three established strategies. Domain randomization [6] randomizes rendering parameters so that the real world falls within the training distribution. Style transfer and image-to-image

translation — CycleGAN and successors [25]— learn a mapping between two image domains from unpaired examples; with only dozens of real target-domain images, this becomes fragile, though patch-based training can partially compensate. Physics- or phenomenology-inspired procedural augmentation — directly simulating the degradation operator on clean images — is the simplest and, where the degradation is well-characterized, the most controllable: annotations are preserved by construction, severity is continuously tunable, and unlimited variants can be generated.

For ALPR, procedural augmentation has an additional appeal: plates are manufactured objects whose degradation process (ink fading toward substrate color, per-glyph wear, surface corrosion) is visually parametrizable. The key open question — and the empirical core of this thesis — is the synthetic-to-real gap: whether a detector trained on procedurally faded plates generalizes to plates whose fading arose from years of physical exposure, with lighting, embossing shadows, and grime that the simulation does not model. Domain-generalization theory and practice both suggest the answer is data- and task-dependent and must be established empirically per task; this study provides that empirical test for the Bangladeshi setting.

8. Detection and Recognition Evaluation Metrics

Detection performance is conventionally reported as precision, recall, and mean average precision (mAP) under the COCO protocol [26] with mAP@0.50 (IoU threshold 0.5) and mAP@0.50:0.95 (averaged over IoU 0.5–0.95) distinguishing coarse from strict localization. PASCAL VOC [27] established the IoU-based

matching protocol these metrics inherit.

Recognition of full plate strings requires sequence-level metrics distinct from detection metrics. The character error rate (CER) is the Levenshtein edit distance [28] between predicted and reference strings normalized by reference length; mean per-image CER must be distinguished from CER over concatenated sequences. Full-sequence exact accuracy — the proportion of plates whose entire string matches exactly — is the deployment-relevant metric and is strictly harsher than CER, since a single wrong character fails the whole plate. When comparing two models on the same test images, paired analyses are appropriate: McNemar's test on per-image exact-match discordance [29] and paired bootstrap confidence intervals resampling images [30]. Treating individual characters as independent observations would violate pairing and inflate significance; this thesis therefore evaluates paired per-image outcomes throughout.

9. Research on Bangladeshi License Plates

Research on Bangla license plates is, as a recent comprehensive study notes, "relatively recent and limited" [14]. Early morphology-based methods achieved approximately 85% accuracy but were sensitive to dirt and color variation; subsequent work introduced CNN-based recognition and the first public Bangladeshi datasets, including the LPDB collection, with character recognition rates around 95% reported for Bangla-specific OCR engines [14]. The current state of the art combines YOLO-family localization with transformer-based text extraction: a YOLO–transformer hybrid reporting 96.5–96.9% detection accuracy and 89.96% end-to-end text recognition on a merged dataset of 40,000 collected

and 70,000 synthetic plates [23]; a YOLOv9-powered Bangla recognition study [31]; and the YOLOv8m + ViT–BanglaBERT framework described in Section 3. A community pipeline fine-tunes YOLOv8 for plate detection with EasyOCR for Bengali text [32]. Public data resources include the Bangladeshi License Plate Recognition Dataset [33] used in this thesis.

Across this body of work, three gaps are visible. First, reported accuracies are overwhelmingly on clean or mildly degraded imagery; weather robustness is typically addressed via fog simulation or restoration pre-processing, not material fading. Second, evaluations are almost exclusively within-dataset; transfer to independently collected, physically degraded plates are rarely tested, and small external test sets are seldom accompanied by manually transcribed ground truth. Third, the interaction between detector post-processing — confidence thresholds, duplicate suppression, ordering — and end-to-end sequence accuracy is reported sparsely, despite its practical importance in detection-based ALPR. This thesis addresses these gaps in part, by comparing detectors across clean, synthetic, and real faded conditions on a small real set.

10. Research Gap

Across this body of work, five gaps are visible. First, evaluation on clean images alone does not establish how well a detector will perform when the visual characteristics of the input change. Robustness should therefore be measured directly on degraded test images.

Second, synthetic data augmentation provides a controlled means of introducing degradation during training, but its benefit depends on the transformation and

the relationship between the synthetic and target domains. Performance on synthetic images cannot be assumed to predict performance on naturally degraded images.

Third, object detection metrics and sequence-recognition metrics measure different aspects of an ALPR system. A detector may localize most characters correctly but still produce an incorrect plate string because of missing detections, duplicate predictions, classification errors, or incorrect ordering.

Fourth, fair comparison requires consistent evaluation protocols. Differences in data splits, checkpoint identities, confidence thresholds, post-processing, and normalization can make apparently comparable results inconsistent.

Finally, evaluating real faded images is difficult when annotations are incomplete. Ground-truth plate strings allow sequence recognition evaluation, but they cannot replace the character-level bounding boxes required to calculate standard object detection metrics (such as precision, recall, and mAP).

Based on these issues, the present study investigates two experiment families using clean and synthetic-fading evaluation conditions and supplements the detection benchmark with recognition evaluation on a separate collection of real faded plates. The design enables the study to examine whether synthetic-fading training improves detection on the selected synthetic degradation and whether corresponding benefits are observed in real-faded sequence recognition.

The contribution is an empirical comparison under the available dataset and evaluation conditions. The study does not claim that synthetic fading solves all adverse-weather problems or that the results establish generalization to every real-world environment. Instead, it seeks to quantify performance differences, assess their consistency, and identify the limits of the available evidence.

Ⅲ. Methodology

This chapter presents the methodology used to investigate the robustness of deep learning-based Bangladeshi license plate detection under image degradation, particularly synthetic fading. The study evaluates whether training object detection models with faded license plate images improves their performance on degraded images compared with models trained on clean images.

The methodology consists of dataset preparation, annotation processing, synthetic fading generation, model training, evaluation on clean and synthetic-faded test sets, and additional recognition experiments using real-faded license plate images. Performance is assessed using object detection metrics and sequence-level recognition metrics. The experimental design follows the general principles of object detection evaluation and robustness testing under image corruption [10], [5].

1. Research Design

Lightweight YOLO detectors read a plate by detecting each character as one of 27 object classes. The manipulation is the training data: clean images only, or clean plus synthetically faded images. Every model is tested on clean images, synthetically faded images, and real faded plates. Two experiment families were run, and their results are never pooled.

Family 1 consists of two YOLOv8n models (Model A, baseline; Model B, trained with BanglaPlateFade, p = 0.60). Family 2 consists of YOLOv5su and YOLO11n, each trained on clean data or on clean plus faded copies. Model A (baseline) is

trained on the clean training images only. Model B (synthetic fading) is trained on the same images, but each image is randomly faded with probability 0.60 at every epoch. All other settings are identical, so comparing A with B isolates the effect of fading during training. In Family 2, the same contrast is expressed by the model names: _clean models are trained on clean images only, and clean_fading models also see faded copies of the training images.

This research adopts a quantitative experimental design. Deep learning object detectors are trained using alternative training datasets and evaluated under clean and degraded image conditions. The principal comparison is between a baseline model trained using clean images and a corresponding model trained using a combination of clean and synthetically faded images.

The research is organized around three experimental components:

- Clean-image detection: Evaluate model performance on images representing the original, relatively clear dataset.
- Synthetic-fading robustness: Evaluate the trained models on a held-out test set containing synthetically faded images.
- Real-faded recognition assessment: Compare predicted license plate character sequences against manually prepared ground-truth transcriptions for a separate set of real-faded plate images.

The first two components measure object detection performance and third measures the correctness of the resulting character sequences. Figure 4 outlines the research workflow. Clean images are preprocessed, optionally faded, and used to train the detectors. The trained models are then tested on clean, synthetic-faded, and real faded images, scored with the metrics that the available labels allow, and compared with paired statistical tests.

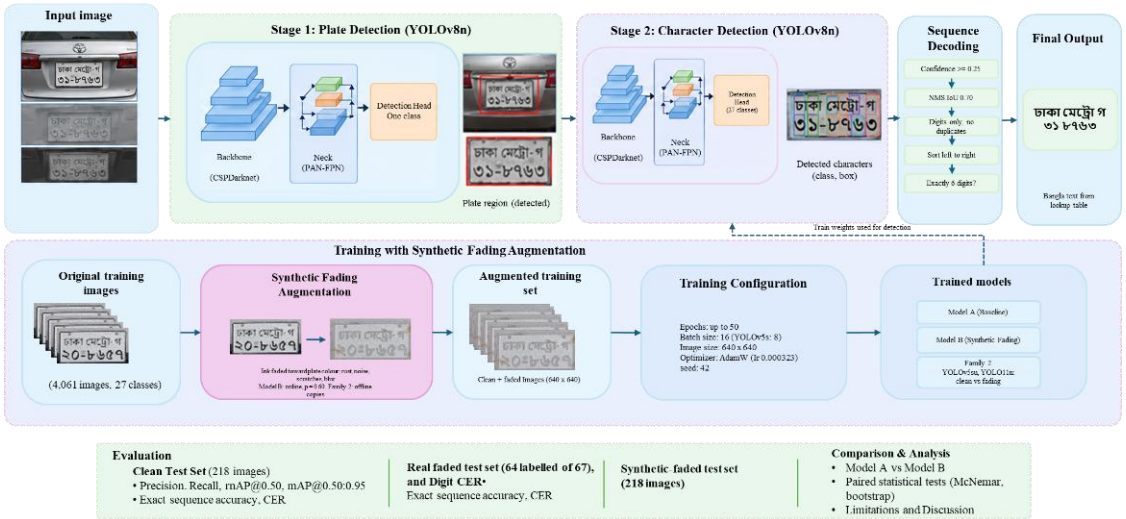


Figure 4: Overall framework for Bangladeshi license plate character detection and recognition with synthetic fading augmentation.

2. Data Set

2.1 Clean dataset

The clean data are Pascal VOC annotations converted to YOLO format, using the splits supplied by the source Kaggle syednahinhossain/bangladeshi-license-plate-recognition-dataset [33].

Table 1 Clean dataset.

Split	Images	Annotated instances
Train	3,480	30,604
Validation	363	3,180
Test	218	1,921

The 27 classes are the digits 0–9, twelve Bengali series letters (ka, kha, ga, gha, ca, ja, ta, da, na, ba, va, ha), and five-word classes (metro, dhaka, khulna, chattogram, jashore). Classes are strongly imbalanced: in training, digit 1 has 4,524 instances and metro 3,192, but na has 32, va 36, ka 44, and da 48.

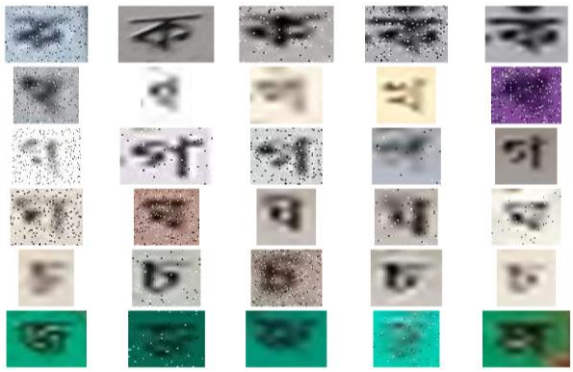


Figure 5: Bengali Character Class Samples

2.2 Synthetic-faded data

Every clean image has one faded copy with identical labels, giving 3,480 + 363 + 218 = 4,061 faded images. For the Family 2 fading models, the 3,480 clean and 3,480 faded training images are combined (6,960); their validation and test sets are the clean ones. A faded image is made from an image of the same split, so fading itself creates no overlap between splits.



Figure 6: Synthetic-faded dataset images

2.3 Real faded plates

The real set has 67 cropped plate images (Figure 7) taken from public datasets [34]- [37]. Sixty-four carry manually transcribed ground truth; three plates were too unclear to transcribe and are excluded.



Figure 7: Real Faded License Plate

Table 2 Summary of the real-faded evaluation set.

Attribute	Value
Total images	67
Images with usable ground-truth transcriptions	64
Images without usable ground-truth transcriptions	3

3. Synthetic Fading

A single function, fade_once, changes pixels only, so labels stay valid. The same code is used offline and online. The complete procedure is called BanglaPlateFade in this thesis. It is used offline to create faded copies of training images (Family 2) and online as a training-time augmentation applied with probability 0.60 (Model B).

Table 3 Operations in fade_once.

Operation	Setting
Ink mask	Otsu threshold clipped to 40–150, then 3×3 opening and closing
Background colour	Median of non-ink pixels brighter than 120
Per-character fade	Strength drawn uniformly from 0.80–0.97 for each connected ink component (0.80–1.0 if under 30 pixels)
Spatial variation	Smoothed random field with values 0.75–1.0
Rust	Blotches, coverage 0.12, colour (155, 100, 65), strongest near plate edges
Noise	Gaussian, σ = 4
Scratches	0–5 grey lines, alpha 30–90
Blur and photometric	Gaussian blur radius 0–1.0; brightness 0.85–1.15; contrast 0.80–1.10

The settings were chosen by visual inspection and were not fitted to the real faded photographs.

A general fading transformation can be as expressed in Eq. (1)

$$\boldsymbol{I_f(x,y) = (1 - w).\ I(x,y) + w.B(x,y)} \quad (1)$$

where:

$\boldsymbol{w}$ = ink mask × fade strength × spatial field

Here:

$\boldsymbol{I(x,y)}$ represents the original image pixel;

$\boldsymbol{I_f(x,y)}$represents the faded pixel;

$\boldsymbol{B(x,y)}$ represents the blending layer;

$\boldsymbol{w}$ controls the overall blending weight, determined by the product of the ink mask, fade strength, and spatial field.

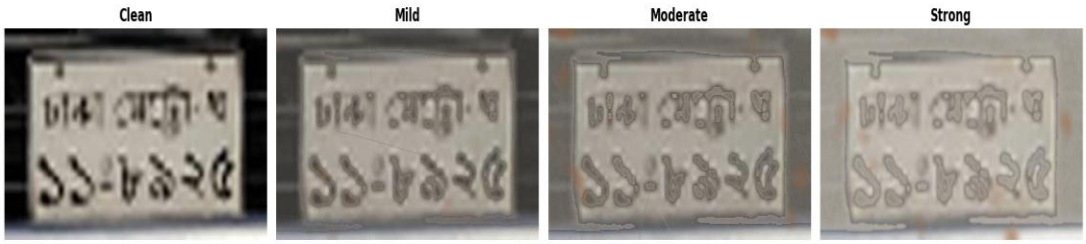


Figure 8: One clean training image and three faded versions

4. Severity levels

Each level is generated from the 218 held-out test images with the fade_once function by changing the fade strength, rust, noise, and blur settings.

Table 4 Fading Severity Levels Used in the Graded-Fading Test

Level	Fade strength	Rust	Noise σ	Max blur radius
Mild	0.35–0.55	0.04	2.0	0.35
Moderate	0.60–0.80	0.08	3.0	0.65
Strong	0.80–0.97	0.12	4.0	1.0

The strong level matches the settings used to generate the training faded images.

5. Models and Training

Best epoch = highest validation fitness (0.1 × mAP@0.50 + 0.9 × mAP@0.50:0.95) on the clean validation split.

Table 5 Trained models.

Model	Architecture	Training data	Batch	Epochs run (best)	Time (h)
Model A	YOLOv8n	Clean (3,480)	16	50 (10)	0.34
Model B	YOLOv8n	Clean + online fading	16	50	2.53
YOLOv5s_clean	YOLOv5su	Clean (3,480)	8	33 (23)	0.41
YOLOv5s_clean_fading	YOLOv5su	Clean + faded (6,960)	8	24 (14)	0.57
YOLO11n_clean	YOLO11n	Clean (3,480)	16	39 (29)	0.32
YOLO11n_clean_fading	YOLO11n	Clean + faded (6,960)	16	24 (14)	0.36

All models start from COCO-pretrained Ultralytics weights and were trained at 640 px, seed 42, with mixed precision and at most 50 epochs. Ultralytics' automatic optimizer choice selected AdamW with learning rate 0.000323. Family 2 used patience 10 and default augmentation, including horizontal flip 0.5; Family 1 used flip-free settings (mosaic 0.5, scale 0.30, translation 0.05, rotation ±2°).

Software: Kaggle – Ultralytics 8.4.174–8.4.175, PyTorch 2.11.0 (CUDA 12.8), Python 3.13, one Tesla T4 GPU.

6. Evaluation Sets and Decoding

Table 6 Evaluation sets.

Set	Used for
Clean test set (218 images)	Precision, recall, mAP@0.50, mAP@0.50:0.95; sequence accuracy, TER, digit CER
Synthetic-faded test set (218 images)	Detection metrics; Models A and B at mild, moderate, and strong fading
Real faded test set (64 labeled of 67)	Strict exact accuracy; digit CER
Comparison and analysis	Paired statistical tests (McNemar, bootstrap); Model A vs Model B; clean vs fading pairs

Detection metrics. Precision, recall, mAP@0.50, and mAP@0.50:0.95 come from the Ultralytics validator on the test split.

Decoding

Detections with confidence ≥ 0.25 were kept. Family 1 kept digit classes, retained the six most confident, and sorted left to right. Family 2 sorted digits left to right (raw) or first applied class-agnostic suppression at IoU ≥ 0.90 (deduplicated). The clean-test decoder split two-line plates into rows.

Precision

Precision measures the proportion of predicted positive detections that are correct as defined in Eq.2

$$P = \frac{TP}{TP+FP} \tag{2}$$

where TP represents true-positive detections and FP represents false-positive detections. Higher precision indicates fewer false-positive predictions relative to the total number of predicted detections.

Recall

Recall measures the proportion of ground-truth objects successfully detected as defined in Eq.3

$$R = \frac{TP}{TP+FN} \tag{3}$$

where FN represents false-negative detections.

Higher recall indicates that a larger proportion of the annotated objects has been detected. Precision and recall provide complementary information about detection performance.

Intersection over Union

Intersection over Union IoU measures the overlap between a predicted bounding box and its corresponding ground-truth bounding box as expressed in Eq.4

$$IoU = \frac{|Bp \cap Bg|}{|Bp \cup Bg|} \tag{4}$$

where Bp is the predicted box and Bg is the ground-truth box.

A predicted box matches a ground-truth box of the same class when their IoU is at least 0.50; mAP@0.50:0.95 averages thresholds from 0.50 to 0.95 in steps of 0.05

Full-Sequence Exact Accuracy

For real-faded recognition, full-sequence exact accuracy measures the

proportion of labeled plates for which the complete predicted string matches the ground truth after predefined normalization. Which defined in Eq.5

$$A_{\text{exact}} = \frac{N_{\text{exact}}}{N_{\text{labeled}}} \times 100 \quad (5)$$

where N_{exact} is the number of exactly correct predicted sequences and N_{labeled} is the total number of labeled test images.

In the strict evaluation, the denominator is 64 labeled images. Missing or incomplete predictions count as incorrect. This avoids inflating the score by excluding difficult images that do not produce a usable prediction.

Character Error Rate

Character Error Rate (CER) measures the minimum number of character-level substitutions, deletions, and insertions required to transform a predicted sequence into its ground-truth sequence, divided by the number of characters in the ground truth as defined in Eq.6

$$CER = \frac{S+D+I}{N} \quad (6)$$

where (S), (D), and (I) denote substitutions, deletions, and insertions, respectively, and (N) is the number of characters in the ground-truth sequence.

For real faded plates, CER is computed per plate on the six-digit registration number: the Levenshtein distance between the predicted and true digit strings divided by 6. A plate with no predicted digits has a CER of 1.0. The reported value is the mean of these per-plate ratios, not the total number of errors divided by the total number of digits. On the clean test set, the digit CER is

computed in the same way on the digit tokens of each image's ordered sequence and averaged over the 218 images.

The token error rate (TER) applies the same idea to the full sequence of class tokens, which includes digits, Bengali series letters, and word classes such as metro and dhaka. For each clean test image it is the token-level edit distance between the predicted and ground-truth sequences divided by the number of ground-truth tokens, averaged over images. Full-sequence exact accuracy requires every token to match; digit-sequence exact accuracy requires every digit to match.

A lower CER indicates better character-level agreement. CER can exceed 1 when the number of edit operations exceeds the ground-truth sequence length.

Table 7 Evaluation metrics and their purposes

Metric	Evaluation stage	Purpose	Preferred direction
Precision	Object detection	Measure correctness of predicted detections	Higher
Recall	Object detection	Measure recovery of ground-truth objects	Higher
mAP50	Object detection	Measure detection quality at IoU 0.50	Higher
mAP50–95	Object detection	Measure detection quality across IoU thresholds	Higher
Full-sequence	Real-faded	Measure completely correct plate	Higher

exact accuracy	recognition	strings	
CER	Real-faded recognition	Measure character-level transcription errors	Lower
TER	Clean-test sequence recognition	Measure token-level transcription errors	Lower

Statistics:

Paired comparisons used an exact two-sided McNemar test (α = 0.05) and a paired percentile bootstrap (20,000 image resamples, seed 42). No multiple-comparison correction was applied.

Ⅳ. Results and Discussion

This chapter presents and discusses the experimental results of the proposed Bangladeshi license plate detection and recognition study. The experiments investigate whether training a YOLO-based detector with synthetically faded license plate images improves its robustness compared with a model trained using clean images.

The evaluation considers three aspects: object detection performance, license plate sequence recognition performance, and recognition performance on a

separate collection of real faded license plate images. Detection performance is assessed using precision, recall, mean Average Precision at an Intersection over Union threshold of 0.50 (mAP@0.50), and mean Average Precision across IoU thresholds from 0.50 to 0.95 (mAP@0.50:0.95). Recognition performance is assessed using complete-sequence accuracy and character-level error measures.

The section first reports the controlled validation results and clean-test performance. It then examines performance on real faded images and discusses the implications, limitations, and reliability of the findings.

1. Experimental Evaluation Overview

Two experiment families were evaluated. Family 1 consists of two YOLOv8n detectors: Model A (baseline), trained on clean images only, and Model B (synthetic fading), trained with online synthetic fading ($p = 0.60$). Family 2 consists of YOLOv5su and YOLO11n, each trained either on clean images only (_clean) or on clean plus synthetically faded copies (_clean_fading). Within each family, the comparison isolates the effect of fading in the training data. The two families are reported separately, because their architectures, augmentation settings, and real-plate decoders differ.

The purpose of this comparison was to determine whether exposure to faded examples during training improves detection and recognition under challenging visual conditions.

The experiments were evaluated on clean test images, synthetic-faded test images, and a separate collection of real faded license plate images. These

evaluation sets serve different purposes. The clean test set measures performance on the original image condition, the synthetic-faded test set measures robustness to generated degradation, and the real-faded set provides an additional test of performance on naturally occurring fading.

2. Object Detection Results

2.1 Validation performance of Model B.

On the 363 validation images, Model B obtained a precision of 94.89%, recall of 89.11%, mAP@0.50 of 94.64%, and mAP@0.50:0.95 of 70.74%. These values were used to select its best checkpoint. A supplementary fine-tuning experiment is reported in Section 4.5.

2.2 Comparison on the Clean Test Set

The main two-model clean-test experiment produced the results shown in Table 8.

Table 8 Object detection performance on the clean test set

Model	Precision (%)	Recall (%)	mAP@0.50 (%)	mAP@0.50:0.95 (%)
Model A – Baseline	94.76	89.18	94.13	70.49
Model B – Synthetic Fading	94.65	92.49	95.70	71.73
Difference (B – A)	−0.11	+3.30	+1.57	+1.24

On the clean test set, Model B achieved a recall of 92.49%, compared with 89.18% for Model A. This represents an increase of approximately 3.30 percentage points. Model B also improved mAP@0.50 by approximately 1.57 percentage points and mAP@0.50:0.95 by approximately 1.24 percentage points.

Precision was slightly lower for Model B, decreasing from 94.76% to 94.65%. The difference was only about 0.11 percentage points. Thus, the synthetic-fading-trained model showed higher recall and mean Average Precision while maintaining a similar precision level.

The increase in recall indicates that Model B detected a greater proportion of the annotated objects under the evaluated conditions. The improvement in mAP suggests that the model also achieved better overall detection performance across the evaluated confidence and localization criteria.

Importantly, these results do not indicate that synthetic fading improved every metric. Instead, the results show a trade-off in which recall and mAP improved slightly while precision remained nearly unchanged.

2.3 Interpretation of Detection Performance

The clean-test results suggest that including synthetic fading during training did not substantially damage performance on clean images. This is an important consideration because a robustness-oriented training strategy should ideally improve performance under degraded conditions without causing a large loss in

normal conditions.

Model B's higher recall is consistent with the possibility that exposure to faded examples helps the detector learn features that remain useful when plate appearance varies. However, this experiment alone does not establish which visual features were learned or prove that synthetic fading is the sole cause of the improvement. Such claims would require controlled repeated experiments or additional ablation studies.

The results should therefore be interpreted as evidence of an observed performance difference between the two recorded models, rather than proof of a general improvement across all datasets or operating environments.

Table 9 Detection performance on the synthetic-faded test set

Model	Condition	Precision	Recall	mAP@0.50	mAP@0.50:0.95
YOLOv5s_clean	Clean	0.937	0.945	0.958	0.716
YOLOv5s_clean	Synthetic-faded	0.759	0.657	0.706	0.502
YOLOv5s_clean_fading	Clean	0.934	0.940	0.967	0.727
YOLOv5s_clean_fading	Synthetic-faded	0.918	0.948	0.961	0.724
YOLO11n_clean	Clean	0.962	0.902	0.949	0.716
YOLO11n_clean	Synthetic-faded	0.742	0.590	0.656	0.467
YOLO11n_clean_fading	Clean	0.916	0.896	0.938	0.706
YOLO11n_clean_fading	Synthetic-faded	0.874	0.913	0.940	0.702

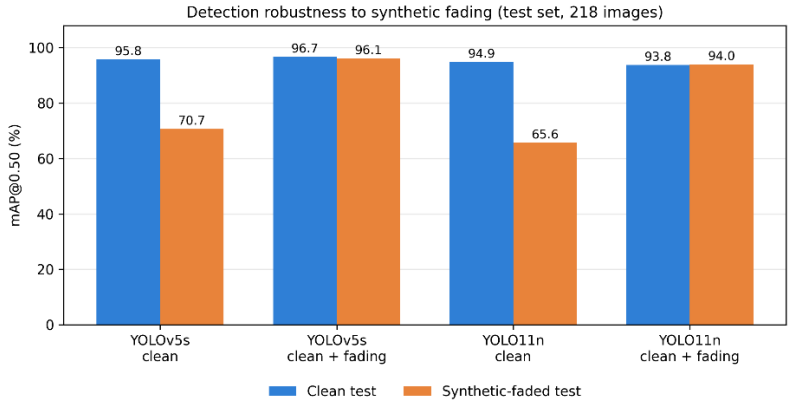


Figure 9: mAP@0.50 on the clean and synthetic-faded test sets for the four models.

- Clean-trained models collapse. From clean to faded images, mAP@0.50 falls by 0.252 (YOLOv5s) and 0.292 (YOLO11n), and recall by 0.288 and 0.312. Most of the loss is missed characters, since precision falls less than recall.
- Fading-trained models hold. Their mAP@0.50 changes by −0.006 (YOLOv5s) and +0.002 (YOLO11n), and mAP@0.50:0.95 by −0.003 and −0.004. On faded images they beat their clean partners by 0.255 and 0.284 mAP@0.50.
- No clear cost on clean images. Fading training changes clean mAP@0.50 by +0.009 (YOLOv5s) and −0.011 (YOLO11n). With one run per model, differences of about 0.01 are not interpretable.

The faded test images come from the same generator as the training fade, so Table 9 shows robustness to this specific degradation, not to fading in general.

3. Models A and B

3.1 Graded synthetic fading

To isolate the effect of the fading augmentation, both models were evaluated at the three severity levels defined in **Error! Reference source not found.** on the same 218 clean test images and on corresponding synthetic variants at mild, moderate, and strong fading severities. The degraded images were generated from held-out test images for evaluation; they were not added to the training data for these test runs. The comparison therefore measures response to the particular synthetic degradation procedure used in this study rather than to all possible real-world conditions.

Table 10 Models A and B on the 218 test images at four fading levels.

Condition	A mAP@50	B mAP@50	Δ mAP@50 (pp)	A Recall	B Recall	Δ Recall (pp)
Clean	94.13	95.70	+1.57	89.18	92.49	+3.30
Mild fading	93.74	95.97	+2.22	88.46	92.97	+4.51
Moderate fading	90.90	95.26	+4.35	82.11	93.29	+11.18
Strong fading	78.39	94.84	+16.44	67.85	90.03	+22.18

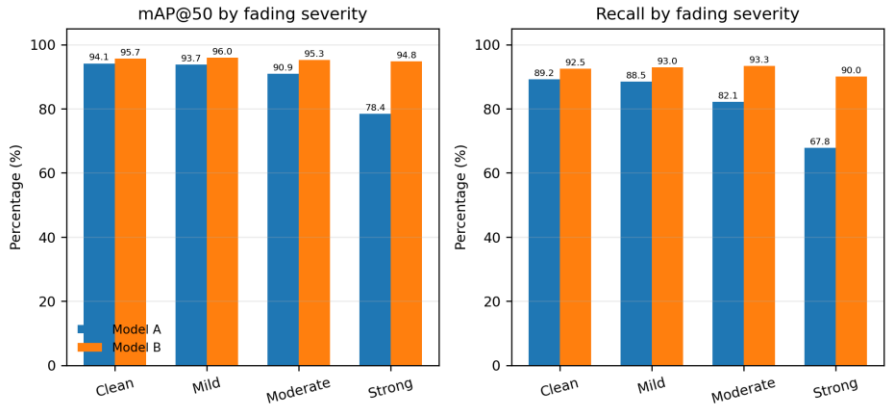


Figure 10: mAP@50 and recall under increasing synthetic fading severity.

On clean images, Model B exceeded Model A by 1.57 points in mAP@50. The gap increased to 2.22 points under mild fading and 4.35 points under moderate fading. Under strong fading, Model A mAP@50 fell to 78.39%, while Model B retained 94.84%, a difference of 16.44 percentage points. Recall showed a similar pattern: at strong severity it was 67.85% for Model A and 90.03% for Model B, a gain of 22.18 points. mAP@50–95 at strong severity was 53.98% for Model A and 69.41% for Model B.

The sharp widening of the performance gap under strong degradation is the strongest evidence that synthetic fading augmentation improved robustness to the fading process represented in the experiments. Model B also had higher mAP@50–95 in the clean, mild, moderate, and strong conditions. Precision under strong fading was 80.90% for Model A and 94.84% for Model B. These observations are consistent with the augmentation helping the model retain features under weakened contrast and simulated surface wear. They do not establish robustness to fog, rain, night lighting, or other degradations that were not separately tested.

3.2 Clean-test sequence recognition

Table 11 Sequence recognition on the 218 clean test images, with paired statistics for B − A.

Metric	Model A %	Model B %	Difference (95% CI)	A-only / B-only correct	McNemar p
Full-sequence exact accuracy (%)	72.48	78.44	+5.96 pp [1.83, 10.09]	5 / 18	0.0106
Digit-sequence exact accuracy (%)	88.07	89.91	+1.83 pp [−1.38, 5.05]	4 / 8	0.388
Token error rate (%)	5.31	3.70	−1.62 pp [−2.69, −0.67]	n/a	n/a
Digit CER (%)	3.28	2.86	−0.42 pp [−1.07, 0.23]	n/a	n/a

On clean images, Model B's advantage is significant for whole-sequence accuracy and token error rate, which include the letters and word classes, but not for the digits alone.

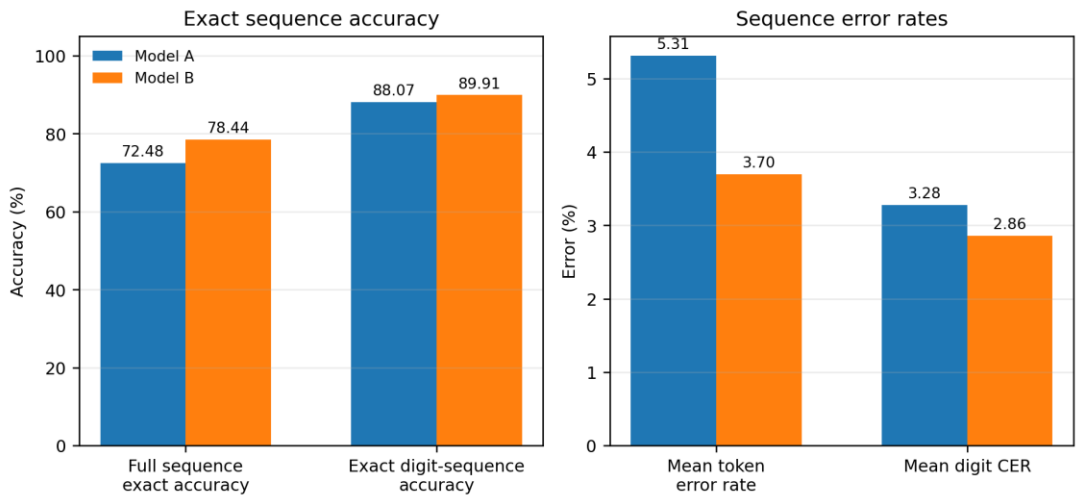


Figure 11: Exact sequence accuracies and average sequence error rates.

4. Real Faded Plates

4.1 Models A and B

Table 12 Registration-number recognition on the 64 labeled real faded plates.

Metric	Model A	Model B
Complete six-digit predictions	54	55
Exact matches	30	43
Strict exact accuracy	46.88%	67.19%
Conditional exact accuracy	55.56%	78.18%
Mean digit CER	18.49%	8.07%

Model B reads 13 more plates exactly (+20.3 percentage points) and cuts the mean digit CER by 10.4 points, a 56% relative reduction. Of the 15 plates on which the models differ, 14 favor Model B and one (plate_0066) favors Model A; 29 plates are correct for both and 20 for neither. The exact McNemar p-value is 0.000977, and a paired bootstrap of the pair counts gives a 95% interval of [9.4, 31.3] points for the accuracy gain. Typical Model A errors are a substituted or dropped digit, for example 111435 for 111335 (plate_0007) or 1658 for 130558 (plate_0015), both read correctly by Model B.

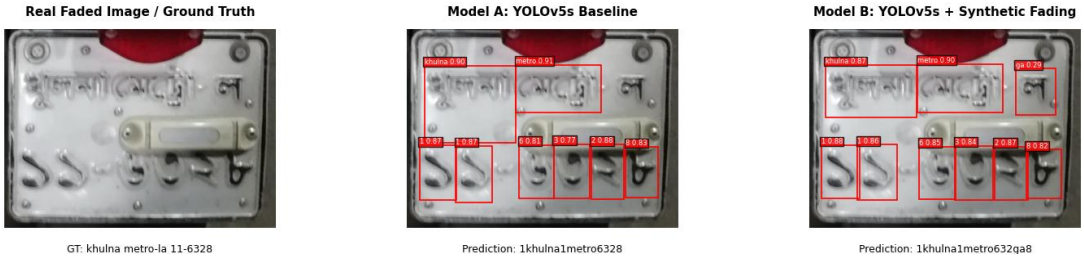


Figure 12: Real faded plates with ground truth and the predictions of Models A and B.

4.2 Family 2: raw and deduplicated decoding

Table 13 Real faded plates, Family 2 (confidence 0.25).

Model	Raw exact	Raw six-digit predictions	Raw mean CER	Deduplicated exact
YOLOv5s_clean	29 (45.3%)	42	20.6%	29 (45.3%)
YOLOv5s_clean_fading	38 (59.4%)	49	13.3%	40 (62.5%)
YOLO11n_clean	28 (43.8%)	38	25.0%	31 (48.4%)
YOLO11n_clean_fading	32 (50.0%)	43	16.4%	37 (57.8%)

Fading training raises raw exact matches by 9 for YOLOv5s and 4 for YOLO11n, and lowers raw CER by 7.3 and 8.6 points. Table 14 shows which of these gains are statistically supported.

Table 14 Paired comparison of raw predictions, fading-trained vs clean partner (64 plates).

Pair	Clean-only / fading-only correct	McNemar p	Exact accuracy difference (95% CI)	CER difference (95% CI)
YOLOv5s	2 / 11	0.0225	+14.1 pp [3.1, 25.0]	−7.3 pp [−12.5, −2.3]
YOLO11n	5 / 9	0.424	+6.3 pp [−4.7, 17.2]	−8.6 pp [−14.1, −3.7]

The YOLOv5s exact-match gain is significant at the 0.05 level; the YOLO11n gain is not, although its CER reduction has an interval that excludes zero.

Duplicate suppression raised exact matches for every model except YOLOv5s_clean: by 2 plates (YOLOv5s_clean_fading), 3 (YOLO11n_clean), and 5 (YOLO11n_clean_fading), and worsened none. The cause is visible in the YOLOv5s_clean_fading detections: 12 plates had more than six digit detections, and 15 box pairs overlapped with IoU ≥ 0.80, usually with different digit labels (for example a 5 and a 0 at IoU 0.99). Class-agnostic suppression cannot remove such pairs, which is why suppression ignores the class. For plate_0037 it turns 1108432 into the correct 110843. Because duplicate suppression (IoU ≥ 0.90) and confidence threshold parameters were evaluated directly on the 64 real faded test plates, these post-processing evaluations are reported strictly as exploratory sensitivity analyses.

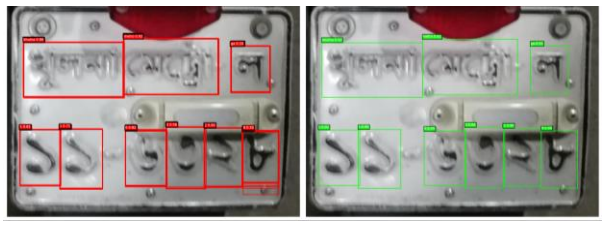


Figure 13: Duplicate digit boxes on a real plate before and after suppression.

4.3 Confidence threshold (YOLOv5s_clean_fading, raw decoding)

Table 15 Effect of the confidence threshold on the 64 real faded plates.

Threshold	Exact matches	Exact accuracy	Mean CER	Six-digit predictions
0.10	38	59.38%	13.28%	49 (76.56%)
0.20	38	59.38%	13.28%	49 (76.56%)
0.25	38	59.38%	13.28%	49 (76.56%)
0.30	36	56.25%	13.02%	50 (78.12%)
0.40	34	53.12%	13.80%	46 (71.88%)
0.50	34	53.12%	15.10%	40 (62.50%)

Exact accuracy is flat up to 0.25 and then falls, while the lowest CER (0.30) and the highest six-digit rate (0.30) occur at a different threshold. The column "six-digit predictions" counts plates with exactly six digits; it is not an accuracy. No threshold was tuned on separate validation data.

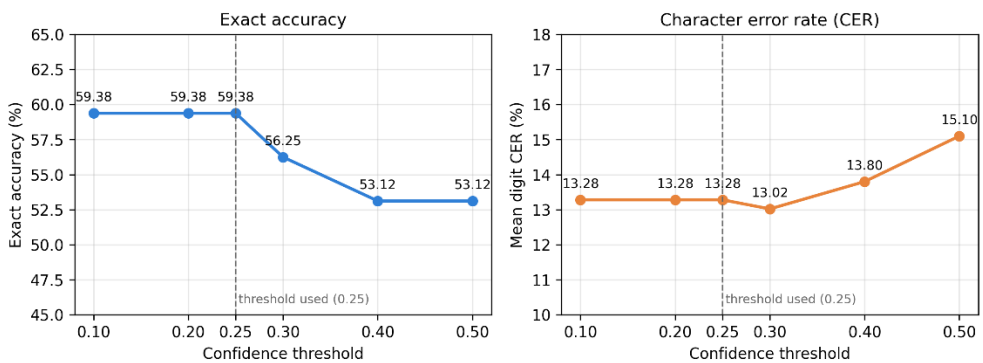


Figure 14: Exact accuracy and CER against confidence threshold.

4.4 Class-wise performance and limitations of rare-class estimates

Overall mAP can conceal differences between individual classes. The class-wise test results show that Model B improved F1 for some character classes,

including ka, na, kha, and ta, while it regressed on da and jashore. Selected results are presented in Table 16. The class support column is important: the clean test set contains only two ka instances, two na instances, five da instances, and nine jashore instances.

Table 16 Selected class-wise test metrics.

Class	Test instances	F1 A (%)	F1 B (%)	Δ F1 (pp)	mAP@50 A (%)	mAP@50 B (%)
ka	2	64.74	83.86	+19.12	51.14	99.50
na	2	60.77	88.75	+27.98	82.83	99.50
da	5	83.31	63.67	-19.64	96.17	69.50
ca	9	78.35	81.76	+3.41	87.79	87.90
jashore	9	98.74	91.31	-7.43	99.50	97.50
kha	10	75.05	88.78	+13.73	86.07	84.43
gha	16	79.25	83.55	+4.30	89.73	89.80
ta	16	92.14	95.90	+3.76	94.38	98.03

Class-wise estimates for these categories are therefore highly sensitive to a small number of predictions and should be interpreted as diagnostic evidence rather than definitive population-level comparisons.

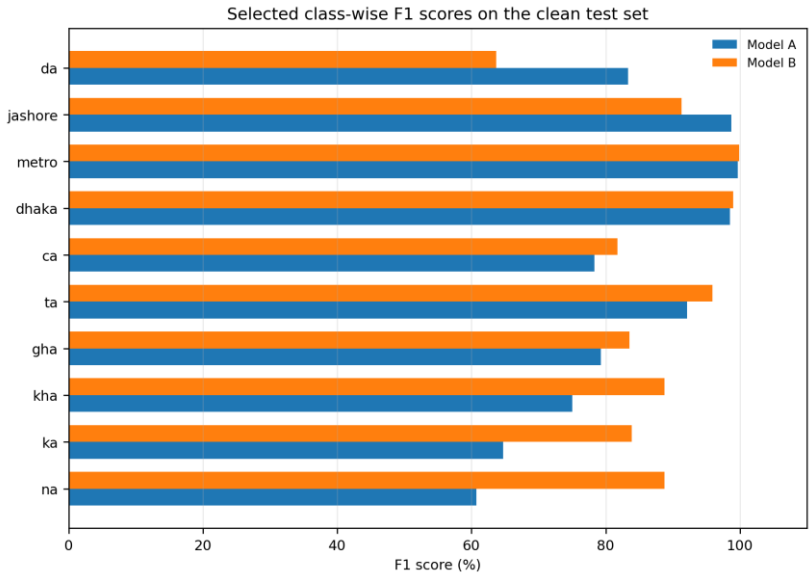


Figure 15: F1 comparison for selected character and word classes.

4.5 Exploratory targeted fine-tuning

The class-wise analysis Table 16 showed that Model B scored lower F1 than Model A on the rare classes da (5 test instances) and jashore (9). Because da has only 48 training instances, a targeted fine-tuning run was tried for this class; jashore was not targeted. The choice was informed by the test-set class-wise results, so the experiment is exploratory, and its outcome is judged on the validation set only.

The 48 training images containing da were each copied twice with random appearance changes, adding 96 images to give 3,576 training images. Model B was fine-tuned from its best checkpoint for up to 25 epochs (AdamW, learning rate 0.0003, batch 16, early stopping with patience 8), with mosaic, mixup, flips, rotation, and translation disabled, scale 0.10, and mild color jitter. Training

stopped after 12 epochs; epoch 4 gave the best validation score.

Here Table 17 compares the original and fine-tuned checkpoints on the 363 validation images.

Table 17 Controlled validation comparison for the exploratory fine-tuned checkpoint

Validation metric	Original Model B (%)	Fine-tuned Model B (%)	Δ (pp)
Precision	94.89	94.63	−0.26
Recall	89.11	91.02	+1.91
mAP@50	94.64	95.29	+0.65
mAP@50–95	70.74	70.63	−0.11

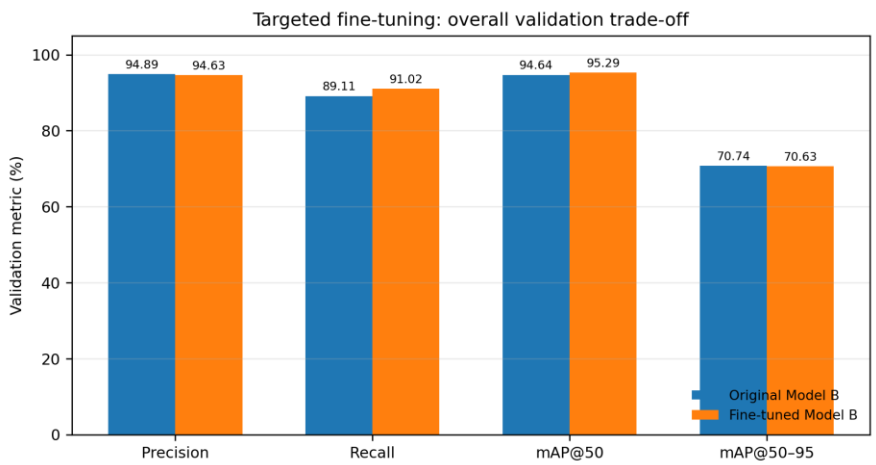


Figure 16: Overall validation trade-off after targeted fine-tuning.

Overall recall rose by 1.91 points and mAP@0.50 by 0.65 points, whereas precision fell by 0.26 points and mAP@0.50:0.95 by 0.11 points. For the target class da, mAP@0.50 rose from 68.25% to 74.50%, but F1 fell from 69.37% to

58.39% because precision dropped from 64.53% to 47.80% while recall stayed at 75.00% (3 of 4 validation instances). For jashore, F1 fell from 88.08% to 85.99%.

The validation set holds only four da instances, so one or two extra false positives are enough to produce these changes. The checkpoint was also selected on the same validation split used for the comparison, so the overall gains are probably optimistic. The fine-tuned model was not used for any test-set or real-plate result, and the targeted objective was not met. The experiment is therefore reported as a negative, exploratory result: rare-class improvement likely needs more real examples, not only augmented copies of the few that exist.

5. Summary of Results

- Synthetic fading removes about a quarter of the clean-trained detectors' mAP@0.50; fading-trained models keep it (Tables 9, 10).
- On clean images, fading training did not clearly cost accuracy, and for Model B it significantly improved whole-sequence accuracy (Table 11).
- On 64 real faded plates, fading training raised exact matches in all three comparisons: Model B over Model A (+20.3 points, $p = 0.001$) and YOLOv5s (+14.1 points, $p = 0.02$) with statistical support, and YOLO11n (+6.3 points) without it (Tables 12–14).
- Duplicate suppression and the confidence threshold change real-plate accuracy by several points, so they are part of the system and must be reported with it (Tables 13, 15).

Ⅴ. Conclusion

1. Introduction

This chapter presents the conclusions of the research on improving the robustness of Bangladeshi vehicle license plate detection and character recognition under degraded visual conditions. The study investigated the effectiveness of synthetic fading augmentation in improving the performance of a lightweight YOLO-based character detection model. The research focused on comparing a baseline model trained without the proposed synthetic fading augmentation against an augmented model trained with the BanglaPlateFade transformation.

The experimental evaluation was conducted using clean test images and a separate collection of real faded license plate images. The analysis considered object detection performance, complete license plate sequence accuracy, character-level error rate, and qualitative detection results. The study also examined the differences between clean, synthetically degraded, and real faded images to understand the limitations of synthetic degradation as a representation of real-world image deterioration.

This chapter summarizes the principal findings, discusses the research contributions, identifies the limitations of the current work, and presents directions for future research.

2. Summary of the Research

The primary objective of this research was to investigate whether synthetic fading augmentation could improve the detection and recognition of Bangladeshi license plate characters under degraded visual conditions.

The research used a lightweight YOLO-based character detection approach. Two experiment families were evaluated. Family 1 consists of two YOLOv8n models: Model A (baseline) and Model B (trained with BanglaPlateFade at p = 0.60). Family 2 consists of YOLOv5su and YOLO11n, each trained with and without faded copies of the training images. The main dataset contained 4,061 images distributed across 27 classes, with 3,480 training images, 363 validation images, and 218 test images. Synthetic fading was incorporated into the training process to expose the augmented model to additional variations in plate appearance.

The evaluation was conducted at two levels. First, the models were compared using object detection metrics, including precision, recall, mean Average Precision at an intersection-over-union threshold of 0.50 (mAP@0.50), and mAP@0.50:0.95. Second, character recognition performance was evaluated using complete sequence accuracy and Character Error Rate (CER).

A separate real-faded image collection was used to investigate how the models performed on degraded plates outside the clean test evaluation. This additional evaluation provided evidence about the practical challenges of recognizing faded characters, although the collection did not contain verified bounding-box annotations for all images.

3. Answers to the Research Questions

RQ1. Clean-trained YOLOv5s and YOLO11n lost 0.252 and 0.292 mAP@0.50; Model A lost 0.157 under strong fading.

RQ2. Yes. Fading-trained models lost at most 0.006 and showed no clear change on clean images (+0.009 and −0.011), though the test fading uses the training generator.

RQ3. Partly. Exact matches rose in all three comparisons (+20.3, +14.1, and +6.3 points), on a small real set.

RQ4. Firmly only for Model B on real plates ($p = 0.001$); the YOLOv5s ($p = 0.0225$) and clean-test sequence ($p = 0.0106$) results are suggestive after correction, and YOLO11n is not significant.

RQ5. Duplicate suppression added 2 to 5 correct plates for three of four models, and raising the threshold from 0.25 to 0.50 lowered exact accuracy from 59.38% to 53.12%.

4. Summary of Major Findings

4.1 Object Detection Performance

The object detection results demonstrated that the synthetic fading model achieved better overall performance than the baseline model on the clean test set.

The baseline model achieved a precision of 94.76%, recall of 89.18%, mAP@0.50 of 94.13%, and mAP@0.50:0.95 of 70.49%. In comparison, the synthetic fading

model achieved a precision of 94.65%, recall of 92.49%, mAP@0.50 of 95.70%, and mAP@0.50:0.95 of 71.73%.

The augmented model therefore achieved higher recall and higher values for both reported mAP metrics, while precision remained approximately the same.
The recall improvement indicates that the augmented model detected a larger proportion of the annotated characters. The increase in mAP suggests that the model achieved better overall detection performance across the evaluated confidence and localization criteria.

These findings indicate that synthetic fading augmentation can be beneficial for training a character detection model. However, because the reported test results were obtained on the clean test set, they should not be interpreted as direct measurements of performance under actual fog, rain, or nighttime conditions.

4.2 Complete License Plate Sequence Accuracy

Complete sequence accuracy was used to evaluate whether the predicted characters formed an entirely correct license plate sequence.
On the 218-image clean test set, Model A achieved an exact sequence accuracy of 72.48%, while Model B achieved 78.44%.

The augmented model therefore improved exact sequence accuracy by approximately 5.96 percentage points.

This finding is important because detecting individual characters correctly does not necessarily guarantee that the complete license plate will be recognized correctly. A single missing, incorrectly classified, or incorrectly ordered character

can cause an entire sequence to be counted as incorrect.

The improvement in complete sequence accuracy suggests that the benefit of augmentation extended beyond the reported object detection metrics to the reconstruction of complete license plate sequences.

4.3 Character Error Rate

On the clean test set, Model A achieved a CER of 3.28%, whereas Model B achieved a CER of 2.86%.

The synthetic fading model reduced CER by approximately 0.42 percentage points. This result indicates that the augmented model made fewer character-level errors relative to the total number of ground-truth characters.

Taken together, the improvements in exact sequence accuracy and CER support the use of synthetic fading augmentation as a potentially effective training strategy for Bangladeshi license plate character detection and sequence reconstruction.

4.4 Evaluation on Real Faded License Plates

The real-faded evaluation provided an additional assessment of model performance on images containing actual plate degradation. Of the 67 images in the collection, 64 had ground-truth labels, while three images were unlabeled and excluded from the sequence-level evaluation.

On the 64 labeled images, Model A achieved an exact sequence accuracy of 46.88%, while Model B achieved 67.19%. The augmented model therefore achieved an improvement of approximately 20.31 percentage points.

The corresponding reported CER values were 18.49% for Model A and 8.07% for Model B.

These results suggest that the synthetic fading model generalized better than the baseline model to the evaluated real-faded image collection. The improvement in exact sequence accuracy and the reduction in CER are consistent with the hypothesis that exposure to synthetic fading during training can help a model recognize characters in degraded plate images.

A second experiment family confirmed the direction of these results. On the synthetic-faded test set, the clean-trained YOLOv5s and YOLO11n lost 0.252 and 0.292 mAP@0.50, whereas their fading-trained counterparts changed by −0.006 and +0.002. On the 64 real faded plates, fading training raised raw exact matches from 29 to 38 for YOLOv5s (+14.1 points, 95% CI 3.1 to 25.0, McNemar $p = 0.0225$) and from 28 to 32 for YOLO11n (+6.3 points, CI −4.7 to 17.2, $p = 0.424$). Only the YOLOv5s gain is statistically supported at the 0.05 level. Across the five exact-match tests reported in this thesis, a Bonferroni threshold is 0.01; only the Model A versus Model B real-plate result ($p = 0.001$) passes it, so the other differences should be read as suggestive.

Nevertheless, the real-faded results should be interpreted cautiously. The evaluation set was relatively small, and it did not have verified character bounding-box ground truth for calculating detection precision, recall, and mAP. Consequently, the results establish a difference in sequence-level recognition performance on this particular collection but do not independently establish robustness across all real-world adverse weather conditions.

5. Research Contributions

The principal contributions of this research are summarized as follows.

First, the study developed and evaluated a synthetic fading augmentation approach for Bangladeshi license plate character detection. The approach was designed to expose a lightweight YOLO-based model to additional variations in plate appearance during training.

Second, the research compared baseline and synthetic fading models using both object detection metrics and sequence-level recognition metrics. This combined evaluation provides a broader assessment than relying exclusively on bounding-box detection performance.

Third, the study evaluated both models on a separate collection of real faded license plate images. The results provide preliminary evidence that synthetic fading augmentation can improve sequence-level recognition on images that differ from the clean test set.

Fourth, the study established an experimental foundation for further research into robust Bangladeshi license plate detection and recognition using lightweight computer vision models.

6. Limitations of the Research

Despite the encouraging results, several limitations should be acknowledged.

6.1 Limited Real-Faded Dataset

The real-faded evaluation contained only 67 images, of which 64 had ground-

truth sequence labels. This sample size limits the extent to which the results can be generalized to the wider population of Bangladeshi vehicle license plates.

The collection may not represent the full variation in plate designs, geographic regions, character conditions, camera distances, illumination, and environmental degradation encountered in practical applications. The real faded plates come from public sources that may overlap with the training data, and no image-overlap check was performed. Clean and synthetic-faded results may therefore be optimistic.

6.2 Limited Annotation for Real-Faded Images

The real-faded collection did not have verified bounding-box annotations for all images. Therefore, the study could not establish real-faded detection mAP, precision, and recall using a complete verified detection ground truth.
Sequence-level metrics remain useful for evaluating recognition outcomes, but they do not independently identify whether errors originated from character localization, classification, character ordering, or sequence reconstruction.

6.3 Synthetic-to-Real Domain Gap

Synthetic fading augmentation cannot be assumed to reproduce every form of real degradation. The observed differences in brightness, contrast, edge strength, and saturation indicate that the synthetic and real-faded image distributions were not identical. The generator models color loss on flat characters. It does not model the embossed relief or paint flaking visible in several real faded plates.
As a result, improvements from synthetic augmentation should not be interpreted

as evidence that all forms of real-world degradation have been addressed.

6.4 Dependence on Character Sequence Reconstruction

Complete sequence accuracy depends on correctly detecting, classifying, and ordering individual characters. Errors in any of these stages can affect the final sequence.

The left-to-right ordering approach used may also require modification for layouts containing multiple rows or special character arrangements. Further work is needed to evaluate the contribution of each stage to the final recognition performance. The confidence threshold and the duplicate-suppression setting were explored on the same 64 plates used for reporting, so the post-processing results are exploratory.

6.5 Generalization and Statistical Uncertainty

The reported improvements are based on the evaluated dataset and experimental configuration. Additional independent test sets and repeated experiments would be useful for assessing variability and statistical uncertainty. The current results should not be interpreted as proof that the augmentation method will produce the same improvement on every dataset or deployment environment.

7. Future Work

- **Expansion of the real-faded dataset:** Collect more real faded Bangladeshi license plate images covering diverse plate designs, regions, viewing angles, physical

wear, and lighting conditions. Include verified plate text, character-level annotations, and separate training, validation, and test sets.

- **More realistic synthetic degradation**: Explore additional degradation techniques, such as uneven illumination, blur, noise, glare, and combined effects. Validate synthetic transformations against real images to improve generalization.
- **Evaluation under fog, rain, and nighttime conditions**: Evaluate model performance under different adverse weather and lighting conditions. Use clearly identified synthetic test data when real images are unavailable, and report results separately for each condition.
- **Confidence threshold optimization**: Analyze how different detection confidence thresholds affect exact sequence accuracy and Character Error Rate (CER). Select optimal thresholds using validation data and evaluate them on an independent test set.
- **Improved character ordering and sequence reconstruction**: Investigate row-aware ordering, spatial relationships, and sequence constraints to reduce errors caused by missing detections, duplicate boxes, and incorrect character ordering.
- **Model efficiency and deployment**: Evaluate model size, inference speed, and computational requirements to determine its suitability for real-time applications and resource-constrained devices.
- **Statistical evaluation and reproducibility**: Conduct repeated experiments, report confidence intervals, and perform appropriate statistical tests. Release configurations, annotation guidelines, evaluation scripts, and documented data splits to support reproducibility.

8. Final Conclusion

This research investigated synthetic fading augmentation for improving Bangladeshi license plate character detection and recognition using a lightweight YOLO-based approach. The baseline model was compared with a model trained using the proposed BanglaPlateFade augmentation.

On the clean test set, the augmented model achieved a higher mAP@0.50, a higher mAP@0.50:0.95, higher recall, and better sequence-level recognition results. Exact sequence accuracy increased from 72.48% to 78.44%, while CER decreased from 3.28% to 2.86%.

On the separate collection of 64 labeled real-faded images, exact sequence accuracy increased from 46.88% to 67.19%, while CER decreased from 18.49% to 8.07%. These findings provide evidence that the augmented model performed better than the baseline on the evaluated real-faded collection.

At the same time, the image-characteristics analysis revealed differences between synthetic and real degradation. The relatively small real-faded dataset and the absence of complete verified character bounding-box annotations also limit the conclusions that can be drawn about real-world detection robustness.

Overall, the findings support synthetic fading augmentation as a promising strategy for improving Bangladeshi license plate recognition under the evaluated conditions. They also demonstrate the importance of assessing both detection and sequence-level recognition performance. Future research should prioritize larger real-world datasets, verified annotations, more realistic degradation models, and explicit evaluation under fog, rain, and nighttime conditions.

The study provides a foundation for further development of robust and

computationally efficient license plate recognition systems for Bangladesh. However, broader claims of adverse-weather robustness require additional experiments using representative real-world data and independently verified evaluation protocols.

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공학석사학위논문

합성 퇴색 증강 기법을 활용한, 저대비·퇴색된 방글라데시 차량 번호판에서의 YOLO 기반 문자 검출 및 시퀀스 인식

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개요

방글라데시 번호판 인식은 퇴색과 낮은 대비로 문자 가시성이 떨어지는 열화된 시각 조건에서 여전히 어려운 과제이다. 본 연구는 경량 YOLO 기반 모델을 사용한 문자 검출과 시퀀스 인식을 개선하기 위해 합성 퇴색 증강을 조사한다.

27개의 문자 및 토큰 클래스를 가진 4,061장의 원본 이미지 데이터셋을 학습 3,480장, 검증 363장, 테스트 218장으로 나누었다. 두 개의 실험군을 수행하였다. 첫 번째 실험군은 YOLOv8n 기준 모델(Model A)과 BanglaPlateFade 기법을 0.60의 온라인 증강 확률로 적용해 학습한 증강 모델(Model B)을 비교하였다. 두 번째 실험군은 합성 퇴색을 사용하거나 사용하지 않고 학습한 YOLOv5su와 YOLO11n을 비교하였다. 성능은 검출 지표, 정확 시퀀스 정확도, 문자 오류율(CER)로 평가하였다. 추가로 실제 퇴색 번호판 이미지 67장을 수집하였으며, 이 중 64장에 시퀀스 수준 평가를 위한 정답 레이블이 있다.

원본 테스트 세트에서 Model B는 mAP@0.50 95.70%, mAP@0.50:0.95 71.73%를 기록하여 Model A의 94.13%와 70.49%보다 높았다. 재현율은 89.18%에서 92.49%로 향상되었고, 정확 시퀀스 정확도는 72.48%에서 78.44%로 증가하였으며 CER은 3.28%에서 2.86%로 감소하였다. 실제 퇴색 데이터셋에서 Model B는 정확 시퀀스 정확도 67.19%, CER 8.07%를 기록하여 각각 46.88%와 18.49%인 Model A보다 우수하였다. 추가 실험에서 합성 퇴색은 YOLOv5s의 정확 일치 수를 14.1%p 향상시켰으나($p = 0.0225$), 이 결과는 Bonferroni 보정 후에는 유의하지 않았다. Model A와 Model B의 차이는 통계적으로 유의하였다($p = 0.001$). 이러한 결과는 평가된 조건에서 합성 퇴색 증강이 방글라데시 번호판 인식을 개선할 가능성을 보여 준다. 다만 실제 퇴색 데이터셋의 규모가 작고 완전히 검증된 바운딩 박스 주석이 없어 실제 환경에서의 검출 견고성에 대한 결론은 제한된다. 향후 연구에서는 더 큰 주석 데이터셋과 안개, 비, 야간 환경을 포함한 추가 열화 조건을 평가해야 한다.

주요어 (Keywords): 방글라데시 번호판 인식, YOLO, 합성 퇴색 증강, 문자 탐지, 정확 시퀀스 일치율, 문자 에러율(CER), 열화된 이미지.

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